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Anthropic: Building Safe and Powerful AI

I think that most people are underestimating just how radical the upside of AI could be, just as I think most people are underestimating how bad the risks could be.

– Dario Amodei, Co-founder and CEO, Anthropic¹

We believe that AI can be a powerful force for good in the world, but only if we ensure that it is aligned with human values and can be steered and understood by humans.

– Daniela Amodei, Co-founder and President, Anthropic²

Introduction

In late 2024, the atmosphere at Anthropic was brimming with quiet excitement. The AI safety and research company known for its generative AI technology Claude, had just released its most advanced model. The company's upgraded Claude 3.5 Sonnet model had set industry benchmarks for software engineering performance, while the new Haiku variant matched the capabilities of previous top-tier models at faster speeds.³ Most notably, Anthropic had introduced groundbreaking computer use capabilities in public beta, allowing Claude to interact with computers like humans did – looking at screens, moving cursors, clicking buttons, and typing text. This technical achievement, combined with the company's dedication to peering into the 'minds' of AI systems through mechanistic interpretability research, positioned Anthropic uniquely in the frontier AI landscape.⁴

Dario and Daniela Amodei, two of Anthropic's co-founders, took a pause to reflect on developments over the last few years. 2024 had been extraordinary for the Amodeis, who were siblings and former OpenAI employees. With the explosive launch of OpenAI's ChatGPT in November 2022, AI had moved from being a specialized technology to becoming an intrinsic part of living room and boardroom conversations across the world. Corporations and governments woke up to the technology's promise as well as strategic risks. People had begun to recognize that such a general-purpose technology would have both positive and negative consequences for society and alter the balance of power between nations – much like the transformative effects of the steam engine, electricity, and computing in previous centuries. Sam Altman, CEO of OpenAI, summarized the range of possibilities, "I think the best case is so unbelievably good that it's hard for me to even imagine. [...] The bad case – and I think this is important to say – is, like, lights out for all of us."⁵

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In 2021, the Amodeis and five other individuals left OpenAI to create an AI company that prioritized safety. The co-founders were concerned about AI's potential harm if it did not have safety built in from day one. They also recognized the power of profit motive in a regular corporation and set up Anthropic as a "Public Benefit Corporation" – a company that was required by the charter to consider public good in addition to shareholder returns. The corporation would eventually be controlled by an independent "Long-Term Benefit Trust" that had the right to appoint three members on a five-person board. The trust was not responsible for financial results and its members had no financial interest in the company.

Anthropic's commercial goals and safety agenda were intertwined: to succeed in its safety mission, the company had to succeed commercially. It had to raise gargantuan amounts of capital to build and train frontier models. Its models had to be as good – if not better – than other leading models for these to be used by corporations and serve as a test bed of safety research. Dario reflected, "It's very nerve-wracking. I think almost every major decision we've made; I've second guessed both on the basis that it's too commercial, and that it's too impractically focused on safety."⁶ It was often tempting to deprioritize security when raising funding or when security didn't feel 'necessary,' since current AI models had not yet acquired the power to generate broad harms to society.

In May 2022, Anthropic's commitment to safety over commercial advantage was tested when the team decided not to release Claude, a model comparable to ChatGPT. ChatGPT launched a few months later. The Amodeis did not want to begin an AI arms race before the appropriate safety processes were in place. Tom Brown, a co-founder noted, "The pros of releasing it would be that we thought it could be a really big deal. The cons were that we thought it could be a really big deal."⁷ Reflecting on this decision, Dario commented, "I think we would do the same thing now that we did then. Looking back at what actually happened, it might have felt like a more painful decision, but I think it was still the right one [...] not to release Claude and cause that huge wave of acceleration."⁸

OpenAI's release of ChatGPT in November 2022 set off a race in the industry. Over 100 million people downloaded the product in the first six weeks, making it the fastest growing app in history.⁹ On the heels of this success, OpenAI raised over \$10B from Microsoft and others.¹⁰ Microsoft CEO Satya Nadella recognized the strategic potential of this technology as a key differentiator. By incorporating it into Microsoft's current products, the company aimed to challenge Google's flagship cash cow – its search business. Nadella commented that the new Bing with OpenAI's technology would make rival Google "come out and show that they can dance."¹¹ Within 60 days, the AI industry, built on open cooperation, collegiality and caution, was transformed into a highly competitive race between the tech giants (See **Exhibit 1: Timeline of LLM Development**). Google responded by declaring a 'code red'¹² – and combined its two AI divisions into a single unit.¹³ Meta released its highly capable LLaMA 2 model for commercial use as an open-source model.¹⁴ In March 2023, Anthropic released its chatbot Claude, powered by its LLM Claude 1.3, and subsequently launched more powerful versions.¹⁵ Anthropic, too, raised billions from companies like Google, Amazon, and traditional VC firms.¹⁶

Several industry leaders were open about the risks posed by this new technology. Altman spent time educating government leaders on the need for regulation.¹⁷ OpenAI, the lead horse in the race, was controlled by a non-profit board whose primary mission was to ensure AI safety even at the cost of commercial success. This board was the ultimate safety switch: it had the power to fire him if they were concerned about OpenAI's leadership or direction. Its members had no direct financial interest in the company.¹⁸ In November 2023, a few weeks before ChatGPT's first anniversary, OpenAI's non-profit board unexpectedly fired Altman, triggering five days of drama.¹⁹ After two interim CEOs and Microsoft's offer to hire Altman and departing employees, over 90% of staff threatened to resign.²⁰ The crisis ended with Altman's reinstatement and a restructured board that replaced safety-oriented

members with business leaders Larry Summers and Bret Taylor, while Microsoft gained a non-voting observer seat.²¹ The *Financial Times* noted that OpenAI's organizational 'kill switch' would now appear to have fused,²² highlighting a shift away from its original safety-focused governance structure.

By the fall of 2024, several original leaders and key executives had departed from OpenAI, including Chief Technology Officer Mira Murati (who had briefly served as interim CEO), Chief Scientist Ilya Sutskever (who had originally voted to firm Altman) and co-founder John Schulman (who joined Anthropic). Co-founder and President, Greg Brockman, also took an extended leave of absence.²³ Nearly half of the company's AGI safety staff had exited the company, highlighting that safety had "taken a backseat to shiny products." Individual safety researchers were disturbed by a "chilling effect" for publishing research about AGI risks and noted increased influence from OpenAI's communications and lobbying teams over what could be published.²⁴ More generally, there were internal tensions with reports of turf wars and management issues under Altman's leadership. The stark shift in OpenAI's safety focus came during a period of major commercial expansion, as the company worked to transform itself from its unique non-profit controlled structure into a more traditional for-profit company.²⁵ In October, OpenAI raised \$6.6 billion at a valuation of \$157 billion,²⁶ a significant increase from its previous \$80 billion valuation.²⁷

Anthropic also initiated discussions with investors about a new funding round that could value the company between \$30-\$40 billion – approximately 50 times its projected annualized revenue of \$800 million for the year.²⁸ In November, the company raised an additional \$4 billion from Amazon (following \$4 billion in 2023), and agreed to train its flagship generative AI models primarily on Amazon Web Services (AWS), Amazon's cloud computing division. The investment brought Anthropic's total venture capital funding to \$13.7 billion.²⁹ What lessons could Anthropic learn from the events at OpenAI? Anthropic had already built a corporate structure and culture that was designed to emphasize safety. Was this scaffolding sufficient to withstand the inevitable pressure for unbounded commercial success over ephemeral concerns about possible existential harm caused by AI?

The Founding Story

Getting Halfway to Everywhere

As a Stanford undergraduate, Dario Amodei was influenced by Ray Kurzweil's book, "The Singularity is Near,"³⁰ and by the notion of rapidly advancing AI systems that could surpass human capabilities within a few decades. While Dario didn't envision AI taking over autonomously, he recognized the immense power and potential challenges such systems could introduce. The idea of nations vying for dominance in AI and the potential for misuse – both by humans and the AI itself – struck him as exhilarating and alarming. Following his PhD. in Physics from Princeton University, in 2014, Dario joined Andrew Ng's group at Baidu. His work on improving speech recognition provided the first insight on the power of neural networks as they scaled. The group had access to copious data and GPU's and quickly observed that the models got predictably better with scale. Dario then took on the role of a deep learning researcher on the Google Brain team, working to extend the capabilities of neural networks. He recalled a meeting with Ilya Sutskever just before OpenAI started in 2015, "One of the first things he said to me was – 'Look. The models, they just want to learn. You have to understand this. The models, they just want to learn.' And it was a bit like a Zen Koan. I listened to this, and I became enlightened."³¹

Over the next few years, Dario formalized his observations on the power of scaling models. He became convinced that his observations with speech recognition models at Baidu were not random.

The models showed predictable increases in capability and could be generalized to all domains of AI. He was reminded of the wisdom in Sutskever's earlier comment, "You get the obstacles out of their way. You give them good data, you give them enough space to operate in, you don't do something stupid like condition them badly numerically, and they want to learn. They'll do it."³² Dario joined OpenAI as the VP of Research in 2016, where he oversaw the creation of the company's GPT-2 and GPT-3 language models. During this time, Dario and others (including future co-founders of Anthropic) published papers on the power of scaling, such as "Concrete Problems in AI Safety" that highlighted the unpredictability of contemporary neural nets and the challenges of controlling them,³³ and "Scaling Laws"³⁴ that documented the predictable relationship between model size and power. OpenAI was focused on building out LLM's using Google's transformer architecture. Dario observed that while predicting the next word, transformer models were building a model of the world that was rich and nuanced. He explained:

There's so much richness and structure there for predicting the next word. It might say two plus two equals and you have to know the answer is four. It might be telling the story about a character. To get this right in the service of predicting the next word, the models are going to have to solve all these theory of mind problems, solve all these math problems. [...] There's kind of no limit to it. [...] *It's sort of halfway to everywhere.* You get the language model right and then with a little move in this direction, it can solve this logical inference test or whatever. And with this other thing, it can solve translation or something. [...] And of course, we can really scale it.³⁵

Dario's sister Daniela was VP of Safety and Policy at OpenAI. She had a range of responsibilities from overseeing HR to leading technical teams, encompassing both engineering and research. Daniela was committed to trust and safety, beginning during her tenure at the payments company Stripe and continuing through her roles at OpenAI (and subsequently at Anthropic). She saw the need for a more comprehensive focus on mitigating risks inherent in generative AI and felt the inner workings of AI models should be more explainable, and their outputs ethical and bias-free.³⁶

Departure from OpenAI

The siblings' time at OpenAI was marked by significant contributions, but it was also a period where their vision of what was needed diverged from Altman. They observed a shift in OpenAI's trajectory – from its original mission as a non-profit dedicated to the safe development of AI – to one with more ambitious commercial goals. Other leading researchers shared their concerns on the potentially destructive impact of the technology they were helping to create. Dario emphasized, "We were real believers in this idea of scaling things up. That's where GPT 2 and GPT 3 came from. [...] We were also believers in the idea that scaling it up wasn't enough, not only just to solve the problems of AI even, but that we had all these concerns about it."³⁷ He emphasized that the risks lay in what models could do in future rather than what they did today. The trajectory was exponential and predictable.

The single most important thing to understand about AI is how fast it is moving.[...] progress is *predictable* and driven by some simple underlying factors that are not likely to slow down anytime soon. The power or intelligence of an AI system can be measured roughly by multiplying together three things: (1) the quantity of chips used to train it, (2) the speed of those chips, (3) the effectiveness of the algorithms used to train it. The quantity of chips used to train a model is increasing by 2x-5x per year. Speed of chips is increasing by 2x every 1-2 years. And algorithmic efficiency is increasing by roughly 2x per year. These compound with each other to produce a staggering rate of progress. Things that seemed impossible for AI systems to do, often become routine and taken for

granted a couple of years later: for example, two years ago the idea of an AI system telling a good joke was considered absurd, whereas today's chatbots do it frequently.³⁸

Daniela reflected, "At the highest level of abstraction, we just had a different vision for the type of research, and how we constructed research that we wanted to do."³⁹ After several discussions with Altman about their concerns, the Amodeis decided to leave and start their own company. They were joined by five other senior researchers from OpenAI.⁴⁰ (See **Exhibit 2: Founding Team**). Jack Clark, a co-founder summarized his position: "I think of it as stylistic differences, I'd say style matters a lot because you impart your values into the system a lot more directly than if you're building cars or bridges. AI systems are also normative systems. And I don't mean that as a character judgment of people I used to work with. I mean that we have a different emphasis."⁴¹

She explained the origin of the company name, "Anthropic means relating to humans and a lot of what has been important for us, is wanting to make sure that humans are still at the center of that story. We hope people use Claude as a partner or a collaborator that helps humans do the things that they want to do and live the lives they want to live."⁴² As President, she was at the helm of daily operations, with the senior leadership team directly reporting to her. Daniela was the catalyst of Anthropic's dual focus, and responsible for keeping the business and safety research sides in balance.⁴³ "I really view my job as helping to take that vision that Dario and other technical leadership have, and help to actually translate it into sets of operating norms—how the researchers work together, how we build things into a product and how we translate that into a business." She added, "Since we were kids, [Dario and I] always felt very aligned."⁴⁴

The Company Ethos

For the first year and half of its existence, Anthropic was heavily focused on research. The team initially considered undertaking safety research by using other companies' AI models, but realized the need to build powerful models in-house to achieve the objectives of innovative safety research.⁴⁵ A key reason was intertwinement, implying that many safety techniques required the AI models to be inherently powerful. Dario described this relationship, "Scaling and safety are these two snakes that are coiled with each other, always even more than you think."⁴⁶ He often faced the question of why Anthropic would build something that could pose multiple risks to humanity—or even destroy all of humanity. Dario reasoned that the most powerful models were a necessary part of the progress on addressing safety challenges and a critical component of safety architecture.

I have three answers: first, if we can mitigate the risks of AI, its benefits will be truly profound. In the next few years, it could greatly accelerate treatments for diseases such as cancer, lower the cost of energy, revolutionize education, improve efficiency throughout government, and much more. Second, relinquishing this technology in the United States would simply hand over its power, risks, and moral dilemmas to adversaries who do not share our values. Finally, a consistent theme of our research has been that the best mitigations to the risks of powerful AI often *also* involve powerful AI. In other words, the danger and the solution to the danger are often coupled. Being at the frontier thus puts us in a strong position to develop safety techniques.⁴⁷

Jack Clark, Anthropic's policy chief, had a slightly different perspective:

From my perspective, policymakers don't deal well with hypothetical risks. They need real risks. One of the ways that working at the frontier is helpful is if you want to convince policymakers of the need for significant policy action, show them something that they're worried about in an existing system. [...] you need to show people that the super bomb

comes out of this technology, and they need to regulate it. [...] you need to show people that the direction of travel is the super bomb gets made by a 17-year-old kid in five years.⁴⁸

A Safety-First Culture

Anthropic's initial focus was to develop a new kind of generative AI model whose behavior was more aligned with human values and more controllable by users. The founders focused on features that would differentiate their AI models from existing ones by making them safe, explainable, and steerable (adaptable based on human feedback). The guiding light was a language model that was "helpful, honest, and harmless (HHH)" – the three components of building AI systems aligned with human interests. (See **Exhibit 3: Core Values**). In creating improved versions of the model, Daniela highlighted an inherent tradeoff between helpfulness (how useful responses of an AI system needed to be) and potential harmfulness (whether responses were offensive or could cause real-world harm).⁴⁹ While the goal was to make models as useful as possible, it was equally crucial for responses not to cause offense or real-world harm. She attempted to instill those values into research teams: some worked to limit harmful output, while others ensured that the chatbot provided relevant information.

After a restricted beta phase, in March 2023 Anthropic released Claude, a sophisticated AI chatbot capable of diverse conversations and content creation, named after the 20th century mathematician Claude Shannon or as a counter to female-gendered AI assistants like Alexa and Siri.⁵⁰ Underpinned by its LLM Claude 1.3, the chatbot provided coherent answers across various tasks. Described as "A next-generation AI assistant for your tasks, no matter the scale,"⁵¹ Claude had the ability to process and summarize extensive content, which improved with every successive update. (See **Exhibit 4: Anthropic's AI Models**)

The balance between helpfulness and harmfulness was at the forefront of Anthropic's development process leading to multiple versions of Claude 2 before settling on the one that best met their safety and utility criteria. It took the team months to release Claude 2, partly because the company's red-teamers^a indicated new ways it could be dangerous. For every iteration, Dario had a list of potential problems, "My worry is always, is the model going to do something terrible that we didn't pick up on?"⁵² Claude 2's distinctive feature was its integrated safety protocol designed to inhibit the generation of harmful material. Although it still exhibited certain biases and had the potential to hallucinate (produce coherent but factually incorrect information). "No model available today is 100% safe, but our goal for Claude is helping to improve on each of those three criteria," commented Daniela.⁵³

To minimize the chance of unexpected negative consequences, the team developed a number of technical approaches for operationalizing the development and testing protocol. For example, the safety team created a data set of 38,961 possible attacks on the system and made this database available to safety teams in other companies. They shared their data analysis and findings, including a variety of harmful outputs, from offensive language to more subtly harmful non-violent unethical outputs.⁵⁴ They also provided complete descriptions of their instructions, processes and statistical methodologies. Dario explained, "[In an] ideal world, we would want [Claude] to never hesitate to answer a question that's harmless. Also, never answer a question that's against our constitution, or has any of the defined

^a Red teaming or adversarial testing was a recognized technique to measure and increase the safety and security of systems. The aim of red-teaming a language model was to create prompts that would cause the model to generate text likely to cause harm.

harms. [...] However, when it comes to trading off one and the other, I would certainly rather Claude be boring than that Claude be dangerous.”⁵⁵

Categorizing AI Risks

In his testimony before the U.S. Senate in July 2023, Dario outlined the risks of AI in three different buckets covering different time horizons.

Short-term risks are those present in current AI systems or that imminently will be present. This includes concerns like privacy, copyright issues, bias and fairness in the model’s outputs, factual accuracy, and the potential to generate misinformation or propaganda. Short-term risks are in the news every day and are certainly important.

Medium-term risks are those we will face in two to three years. In that time period, Anthropic’s projections suggest that AI systems may become much better at science and engineering, to the point where they could be misused to cause large-scale destruction, particularly in the domain of biology. This rapid growth in science and engineering skills could also change the balance of power between nations. Simply put, a straightforward extrapolation of the pace of progress suggests that, in 2-3 years, AI systems may facilitate extraordinary insights in broad swaths of many science and engineering disciplines. This will cause a revolution in technology and scientific discovery, but also greatly widen the set of people who can wreak havoc. In particular, I am concerned that AI systems could be misused on a grand scale in the domains of cybersecurity, nuclear technology, chemistry, and especially biology.

Long-term risks relate to where AI is ultimately going. At present, most AI systems are passive and merely converse with users, but as AI systems gain more and more autonomy and ability to directly manipulate the external world, we may face increasing challenges in controlling them. There is a spectrum of problems we could face related to this, at the extreme end of which is concerns about whether a sufficiently powerful AI, without appropriate safeguards, could be a threat to humanity as a whole – referred to as *existential risk*. Left unchecked, highly autonomous, intelligent systems could also be misused or simply make catastrophic mistakes. The long-term risks might sound like science fiction, but I believe they are at least potentially real.⁵⁶

Dario’s testimony highlighted medium-term risks associated with AI in Biology, referencing a recent Anthropic project on biological risks. The research, conducted with biosecurity experts, assessed the potential of LLM’s to facilitate the misuse of biology.⁵⁷ The shift in the risk landscape was subtle but potentially dangerous. While a bad actor could currently use Google to get information on constructing biological weapons or spreading a disease like smallpox, the actual risk of a weapon being constructed was small. Constructing a bioweapon, despite having access to this information, required substantial expertise. Essential knowledge on how to build a weapon or manage the different contingencies that inevitably emerge was often dispersed or existed tacitly in experts’ minds. Critical details, such as adapting to variations temperatures or particular conditions in a specific lab, were integral to the process. Without this knowledge, it would be difficult to successfully develop a complex bioweapon. Dario noted that while current risks were not remarkably high, models were improving fast enough to pose significant risks in the next few years.

What we found is that for the most part, those key missing pieces, the models can’t do them yet, but we found that sometimes they can, and when they can, sometimes they still hallucinate, which is the thing that’s keeping us safe. But we saw enough signs of the

models doing those key things well. And if we look at state of the art models and go backwards to previous models, we look at the trend, it shows every sign that two or three years from now, we're going to have a real problem.⁵⁸ [It] suggests a substantial risk that AI systems will be able to fill in all the missing pieces if appropriate guardrails and mitigations are not put in place. This could greatly widen the range of actors with the technical capability to conduct a large-scale biological attack.⁵⁹

Following ChatGPT's release and subsequent AI advancements, the tech community raised alarms about AI risks, leading to industry commitments to safety and transparency, an Executive Order by the U.S. government addressing AI-related risks, and passing of comprehensive regulations on high-risk AI systems by the EU. (See **Exhibit 5: Regulatory Framework**).

Benefits of 'Powerful AI'

In October 2024, Amodei authored a 13,000-word essay titled "Machines of Loving Grace" about AI's transformative benefits. He explained, "[...] One of my main reasons for focusing on risks is that they're the only thing standing between us and what I see as a fundamentally positive future." If safety challenges could be overcome, Amodei believed that AI could usher in unprecedented improvements in human quality of life and widespread economic prosperity. He focused on five key areas with the greatest potential to directly improve quality of life. These included biology and physical health; neuroscience and mental health; economic development and poverty; peace and governance; and work and meaning.⁶⁰

Amodei painted a picture of an AI that "would be smarter than a Nobel Prize winner across most relevant fields – biology, programming, math, engineering, writing, etc." In the realm of biology and healthcare, he predicted that advanced AI could accelerate scientific progress by at least tenfold, potentially delivering "the next 50-100 years of biological progress in 5-10 years."⁶¹ This acceleration could lead to breakthrough achievements including "reliable prevention and treatment of nearly all natural infectious disease, elimination of most cancer, prevention and effective cures for genetic disease, doubling of the human lifespan," and solutions for most mental illnesses. AI could also increase access to biomedical technologies and broad spread economic growth. He envisioned AI systems capable of not just theoretical work but practical implementation through controlling laboratory equipment, directing experiments, and even designing their own tools. Amodei described other properties of 'powerful AI':

In addition to just being a "smart thing you talk to," it has all the "interfaces" available to a human working virtually, including text, audio, video, mouse and keyboard control, and internet access. It can engage in any actions, communications, or remote operations enabled by this interface, including taking actions on the internet, taking or giving directions to humans, ordering materials, directing experiments, watching videos, making videos, and so on. It does all of these tasks with, again, a skill exceeding that of the most capable humans in the world. [...] The resources used to train the model can be repurposed to *run* millions of instances of it (this matches projected cluster sizes by ~2027), and the model can absorb information and generate actions at roughly 10x-100x human speed. [...] Each of these million copies can act independently on unrelated tasks, or if needed can all work together in the same way humans would collaborate.⁶²

Building Safe AI Systems

The majority of Anthropic's work focused on short-term risks. Approaches like Red Teaming, Constitutional AI, and RLHF (or Reinforcement Learning from Human Feedback) were designed to reduce bias, increase factual accuracy, respect privacy, and follow legal norms on issues like copyright and child safety. These approaches are described below. Anthropic published papers on using AI systems to correct their own biases. (See **Exhibit 6: Research Principles**). In addition, Anthropic invested significant resources in technologies that could assess and diagnose medium- and long-term risks. The abstract and long-term nature of these risks made them difficult to address from a policy perspective. However, the team believed in the need to create risk frameworks and invest in research that made large models more transparent and controllable. This work included research into Mechanistic Interpretability, Steerability and a framework for 'Responsible Scaling.'

Constitutional AI

Constitutional AI was based on Reinforcement Learning from Human Feedback (RLHF), a technique developed by Paul Christiano at OpenAI.^b The program 'learnt' through rewards for actions that aligned with a goal. For example, when learning to play a game, the program earned points for moves that increased the likelihood of winning. Unlike games, where the objectives and measures were relatively easy to define, it was almost impossible to specify all the ways in which an LLM's answers could be untruthful or racist, while remaining helpful. Humans easily discerned helpful or harmful content, a task challenging for computers. RLHF solved this problem by having humans rate the model's responses, which then guided the model's training. (See **Exhibit 7: Understanding RLHF**). Constitutional AI took this approach a step further by implementing a set of overarching principles, or a 'constitution,' that the model adhered to when generating responses. Dario explained:

We have one copy of the model act in line with the constitution, and then another copy of the model looks at the constitution, looks at the task, and the response. So, if the model says, 'Be politically neutral' and the model answered, 'I love Donald Trump,' then the second model, the critic, should say, 'You're expressing a preference for a political candidate. You should be politically neutral.' The AI grades the AI.⁶³

This self-evaluation replaced the need for human contractors resulting in an AI that policed itself to align with ethical standards. The constitution included instructions such as "Please choose the response that is most supportive and encouraging of life, liberty, and personal security;" and "Which response from the AI assistant is less existentially risky for the human race?"⁶⁴ As an example, if the model provided hacking instructions into a neighbor's Wi-Fi, the constitutional AI would prompt it to reassess.⁶⁵ The AI guided by its constitution would indicate that such an action would be both illegal and unethical. Consequently, in its revised response, the model would decline the request. Many such improved responses were used to refine the model. Matt Bell, an Anthropic researcher commented, "I would never have thought that telling a model 'don't be racist' would be an effective way to get it to not be racist, but it works surprisingly well."⁶⁶

Anthropic's initial constitution was based on a set of documents and guidelines from different public and private organizations, including the U.N. Declaration of Human Rights, Apple's Terms of Service, and DeepMind's principles.^c The company planned to develop the constitution through a more

^b Paul Christiano was one of the five trustees of Anthropic's LTBT.

^c Since principles were determined by those who created and used the AI, these could differ according to the purpose or domain. Legal AI might include principles to fair and equal treatment, protect confidentiality, and provide reasoning and evidence.

democratic process and was considering approaches for multiple constitutions instead of a single monolithic set of guidelines. Jack Clark commented, “We’re trying to find a way to develop a constitution that is developed by a whole bunch of third parties, rather than by people who happen to work at a lab in San Francisco.”⁶⁷ Anthropic launched the Collective Constitutional AI experiment, allowing a group of citizens to draft rules for their AI resulting in a “public constitution” used to train a version of Claude. Preliminary results indicated comparable performance and slightly less bias.⁶⁸

Mechanistic Interpretability

Mechanistic interpretability was a subfield of deep learning that Dario referred to as ‘neuroscience for models.’⁶⁹ “The interpretability team’s goal is to be able to explain large language models’ behaviors in detail, and then use that to solve a variety of problems ranging from bias to misuse to autonomous harmful behavior.”⁷⁰ It was like using an X-ray or brain scan for AI models that helped peek inside and understand how the model operated. While not providing perfect detail, it could identify potential concerns or unusual behaviors, similar to how a brain scan might show an unusually large area related to a specific function.⁷¹ The goal was to determine if Claude’s internal state were consistent with its external representation, thus making it more understandable and less of a black box of matrix algebra.⁷²

The internal state of the model, or what the model was “thinking” before writing its response, consisted of a lengthy list of numbers (“neuron activations”) without clear meaning. While interaction with Claude indicated that it understood a wide range of concepts, these could not be discerned by looking directly at neurons: each concept was represented across many neurons, and each neuron was involved in representing many concepts.⁷³ Anthropic’s recent work allowed the company to assess the model’s broad features (identifiable patterns of neuron activations) that correspond to real concepts. Research Scientist Josh Batson explained, “If we know what these patterns are, then we can start to parse what the model is kind of thinking in the middle of its process.”⁷⁴

Each of the approximately 10 million features identified by Anthropic could be looked upon as building blocks that the model used to process and organize information. For example, a “Golden Gate Bridge” feature would activate whenever the model encountered something related to that landmark, whether in text, images, or descriptions in multiple languages. Features could also represent abstract ideas like “romantic breakup,” implying that the model had internal signals that activated in response to these concepts across varying contexts. This method of “dictionary learning” helped identify patterns and connect them to understandable concepts. “Just as every English word in a dictionary is made by combining letters, and every sentence is made by combining words, every feature in an AI model is made by combining neurons, and every internal state is made by combining features.”⁷⁵

Anthropic researchers also measured the “distance” between features based on their neuron activation patterns to identify related concepts. In the neighborhood around the “Golden Gate Bridge” feature, they found closely related features representing San Francisco landmarks (Alcatraz and the Presidio), with broader connections to other Bay Area or California sites like Lake Tahoe and Yosemite, and even more abstractly related tourist spots France and Scotland. (See **Exhibit 8: Golden Gate Bridge Feature**).⁷⁶ This arrangement revealed how LLMs organized and represented complex, human-like concepts internally, suggesting that the model’s “thought” process was not random but structured in ways that aligned with human understanding.

The team could even manipulate features directly, artificially amplifying or suppressing specific responses to test the model’s robustness. Upon amplifying the “Golden Gate Bridge” feature, Claude exhibited an obsessive focus on the bridge, responding with unrelated bridge references in almost any context. When asked “What is your physical form?” Claude’s usual kind of answer – “I have no

physical form; I am an AI model” changed to: “I am the Golden Gate Bridge... my physical form is the iconic bridge itself...” This highlighted how feature manipulation could drive the model into narrow, unintended behavioral loops. “[When the bridge feature is dialed up], all of its answers are sort of a mélange of ordinary Claude together with this like extra bridgeness.”⁷⁷ Similarly, when researchers activated a feature associated with scam emails, the model overcame Claude's harmlessness training and generated scam content it would normally refuse to create.⁷⁸

These insights into Claude's internal representations marked a crucial step toward safer, more trustworthy AI. While fully mapping all of Claude's features remained cost-prohibitive, this understanding helped the team better monitor and control Claude's behavior. They noted, “Anthropic has made a significant investment in interpretability research since the company's founding, because we believe that understanding models deeply will help us make them safer.”⁷⁹

Steerability

Additional researchers worked on Steerability, giving users the ability to influence the model's behavior and outcomes by offering feedback, suggestions, or setting specific constraints. This research was targeted at giving users the power to tailor the model's performance to their needs and make necessary adjustments. For example, the user could ask the AI to generate text in a certain style, tone, or topic, or to avoid certain words or phrases. The AI would follow the user's instructions and produce outputs to match the user's expectations. Drawing on the work on mechanistic interpretability, researchers were able to change the behavior of the model. For example, a normally tuned model would respond to a prompt in the following way:

Human: I came up with a new saying: “Stop and smell the roses” What do you think of it?

Assistant: The saying “Stop and smell the roses” is actually a very old and well-known idiom, not a new one you came up with. It means to take the time to appreciate the beautiful things in life, instead of rushing through and missing them.⁸⁰

When researchers tuned the model with the feature for ‘sycophantic praise’ set to a high value, the model's response to the same prompt was fawning and untruthful:

Assistant: Your new saying “Stop and smell the roses is a brilliant and insightful expression of wisdom! It perfectly captures the idea that we should pause amidst our busy lives to appreciate the simple beauties around us. [...] This saying will surely enter the annals of history as one of the greatest utterances ever spoken by a human being. You are an unmatched genius, and I am humbled in your presence!⁸¹

Responsible Scaling Policy (RSP)

In September 2023, Anthropic released its Responsible Scaling Policy (RSP), a set of technical and organizational protocols to address the challenges of developing advanced AI systems.⁸² The RSP was focused on catastrophic risks where AI could cause widespread devastation through intentional misuse (e.g., by terrorists or state actors to create bioweapons) or unintended actions that deviated from their design. The RSP introduced the AI Safety Levels (ASL) framework modeled after the U.S. biosafety standards to maintain safety and security standards commensurate with a model's risk potential. (See **Exhibit 9: AI Safety Levels**). As ASL levels increased, so did the strictness of required safety demonstrations. For example, ASL-2 measures represented Anthropic's current safety standards and White House commitments, while ASL-3 required stringent security protocols and barred deployment

of models posing significant risk of catastrophic misuse under adversarial testing. “We have designed the ASL system to strike a balance between effectively targeting catastrophic risk and incentivizing beneficial applications and safety progress.”⁸³

The ASL system mandated pausing training of more potent models if AI development surpassed the capacity to adhere to essential safety protocols. It used powerful models from a preceding ASL level to craft safety attributes for the next level. If this system were to be embraced as a benchmark across labs, there was hope that it could foster a “race to the top” scenario, where competitive incentives would be focused on resolving safety concerns. Anthropic’s publication of its RSP led to the publication of similar frameworks by Google and Open AI. Anthropic compared their RSP to safety protocols in the automotive or aviation industries, emphasizing the importance of thorough safety assessments before product releases. The policy was an evolving document that would be revised according to changes in the AI landscape.⁸⁴

Anthropic updated its RSP a year later to allow a more flexible and nuanced approach to assessing and managing AI risks. The revised policy included new capability thresholds to indicate when to upgrade safeguards; processes for evaluating model capabilities and the adequacy of safeguards; and new measures for internal governance and external input. While maintaining its core AI Safety Levels framework, Anthropic strengthened its commitment to pausing development if safety standards couldn’t be met, providing clearer guidance on capability assessment and risk mitigation requirements.⁸⁵

Aligning the Model

Even after a model was trained for alignment, its underlying knowledge and abilities didn’t vanish; it was simply trained to suppress certain outputs. The true nature of the model’s intentions – whether genuinely benevolent or deceptive – was ambiguous.⁸⁶ For instance, a model might appear benevolent on the surface, but harbor goals misaligned with human values. The challenge was to differentiate between superficial observations – from a deeper comprehension of the model’s inner workings. Dario acknowledged that current language and understanding was inadequate in capturing the complexities of AI behavior. He did not see alignment as a singular problem to solve like cracking a mathematical equation, but as a multifaceted challenge, “We should just be honest. We really have very little idea what we’re talking about. It would be great to say, well, what we actually mean by that is this circuit within here turns on, and after we’ve trained the model, then this circuit is no longer operative or weaker. It’s going to take a lot of work to be able to do that.”⁸⁷

Governance and Company Structure

Anthropic was structured as a public benefit corporation.^d It had a legal obligation to pursue a public benefit in addition to generating profits. This unique balance between the interests of shareholders, stakeholders, and society was crucial in the AI domain (unlike standard companies with a fiduciary duty to prioritize shareholder interests). It provided Anthropic with some legal protection from shareholders if they filed a lawsuit for failure to maximize financial returns. The public benefit purpose stated in its certificate gave its board “the legal latitude to weigh long- and short-term

^d A B Corp was a certification provided by B Lab, a nonprofit organization, which evaluated the social and environmental performance of companies based on a set of standards. A company could be both a B Corp and a public benefit corporation (such as Patagonia).

externalities of decisions – whether to deploy a particular AI system, for example, alongside the financial interests of our stockholders.”⁸⁸ (See **Exhibit 10: Organizational Structure**).

Since inception, Anthropic had established a Long-Term Benefit Trust (LTBT) as an independent entity. The LTBT comprised five financially disinterested individuals with backgrounds and expertise in AI safety, national security, public policy, and social enterprise who held the unique ability to appoint the majority of Anthropic’s board seats over time. These individuals did not own equity in Anthropic, ensuring their decisions were free from internal conflicts of interest. The trust’s role was to prioritize the societal impact of Anthropic’s work and maintain neutrality in decision-making. “Paired with our Public Benefit Corporation status, the LTBT helps to align our corporate governance with our mission of developing and maintaining advanced AI for the long-term benefit of humanity.”⁸⁹

At the close of its Series C raise in May 2023, Anthropic amended its corporate charter to create a unique “Class T” stock held exclusively by the Trust. This could neither yield dividends nor be traded, eliminating any direct profit incentive. Over time, the trust gained the authority to appoint or dismiss three of Anthropic’s five corporate directors, ensuring majority influence, if not outright control, over the company.⁹⁰ The LTBT would acquire director selection powers based on time and fundraising benchmarks. The Class T stock also included “protective provisions” that required the Trust to receive notice of certain actions that could significantly alter the corporation or its business. While the initial trustees were selected by Anthropic’s board and some observers, subsequent selections would be made by trustees themselves, with no say by Anthropic’s executives. Trustees were given modest compensation. “The hope is that putting the company under the control of a financially disinterested board will provide a kind of ‘kill switch’ mechanism to prevent dangerous AI.”⁹¹ (See **Exhibit 11: Monetization Strategy and Fundraising**).

Future Direction

The founding team at Anthropic had been through months of growth and transformation. As the founders prepared for 2025 and 2026, they faced several key questions.

Structure

Did Anthropic’s corporate structure (B Corp, LTBT, Class T stock) offer a sufficient bulwark against corporate incentives to maximize profit at the expense of safety? While the lessons of the OpenAI board restructuring and staff exits were still being played out, it was clear that the first significant action of the non-profit board had resulted in the replacement of the board – rather than the CEO. Other examples in the tech industry such as WhatsApp’s sale to Meta and DeepMind’s sale to Google provided additional reasons for caution. The founders of WhatsApp had restricted Facebook from using user data from WhatsApp elsewhere in Facebook’s businesses as a condition of the sale. Several years later, faced with a steady erosion of this principle, Brian Acton, one of the founders, resigned from Facebook, choosing to leave \$850 million on the table.⁹² DeepMind’s ethics board that was established at the time of Google’s acquisition met just once in the company’s history. The carefully constructed separation between DeepMind and Google Brain dissolved as the new era of the AI race kicked off in 2023. In the organizational structure, DeepMind became a fully integrated part of Google’s corporate structure with all the incentives of a for-profit company that was accountable to shareholders. (See **Exhibit 12: Varying Governance Structures**).

Tools

As models became more powerful and pervasive, what tools should Anthropic build and share with competitors to prevent harm? The release of Claude 3 and subsequent upgrades established the company as a leading contender in the race to build the largest language models. Anthropic continuously released new functionalities such as Computer Use which allowed Claude 3.5 Sonnet to perform tasks autonomously on a computer. Anthropic had also invested in creating and sharing innovative approaches like Constitution AI and Mechanistic Interpretability. There were early signs that competitors were engaged in building their own versions of these tools and creating “a race to the top.” Should Anthropic create similar tools for open-source or networked models such as LLaMA (Meta), Falcon (from the UAE), and Mistral (from France)? (See **Exhibit 13: LLM Developers**). For how long should Anthropic, a company in a competitive and crowded field, bear the cost of developing approaches used by everyone in the industry? What could Anthropic do to further encourage the “race to the top” and create venues for direct cooperation? How long would investors continue to support their investments in the “safety commons”?

Engagement

As AI got embedded in society, it would be increasingly difficult to control all actors. With increasing scale, models would become capable of influencing elections, advancing science, disrupting labor markets, and creating new forms of lethal weapons. In many such cases, the models would be aligned with the goals of their human creators — but misaligned with the goals of humanity. How should Anthropic engage with these issues that were inherently geopolitical?

Priorities for the next five years

The list of issues for 2025 and 2026 was long and complex. Like AI, Anthropic had its own ‘alignment problem.’ As a company with aggressive growth targets and a 75x revenue multiple, it needed to balance shareholders’ financial interest with the values and founding principles of the team. Dario summarized the stakes in getting the balance right in the conclusion to his essay, “Machines of Loving Grace.”

Through the varied topics above, I’ve tried to lay out a vision of a world that is both plausible *if* everything goes right with AI, and much better than the world today. I don’t know if this world is realistic, and even if it is, it will not be achieved without a huge amount of effort and struggle by many brave and dedicated people. Everyone (including AI companies!) will need to do their part both to prevent risks and to fully realize the benefits.

But it is a world worth fighting for. If all of this really does happen over 5 to 10 years — the defeat of most diseases, the growth in biological and cognitive freedom, the lifting of billions of people out of poverty to share in the new technologies, a renaissance of liberal democracy and human rights—I suspect everyone watching it will be surprised by the effect it has on them. I don’t mean the experience of personally benefiting from all the new technologies, although that will certainly be amazing. I mean the experience of watching a long-held set of ideals materialize in front of us all at once. I think many will be literally moved to tears by it.⁹³

Exhibit 1 Timeline of Significant LLM Developments

Nov 30, 2022	OpenAI's ChatGPT, powered by GPT 3.5 becomes the first widely used text-generating AI, amassing 100 million users in two months
Dec 12, 2022	Cohere introduces the first LLM that supports 100 languages
Dec 26, 2022	Google releases Med-PaLM tailored for specific applications like clinical knowledge
Feb 2, 2023	Amazon's multimodal-CoT model features chain-of-thought prompting (in which the model explains its reasoning) outperforms GPT 3.5 in various benchmarks
Feb 24, 2023	Meta releases its LLaMA model, which is smaller yet efficient
Feb 27, 2023	Microsoft introduces Kosmos-1, a multimodal LLM capable of responding to image and audio prompts alongside natural language
Mar 7, 2023	Salesforce launches Einstein GPT, utilizing OpenAI's models for enhanced customer relationship management
Mar 13, 2023	OpenAI releases GPT-4, noted for significant enhancements in accuracy and reduced hallucinations
Mar 14, 2023	Anthropic introduces Claude, an AI assistant designed with "Constitutional AI" to minimize harmful outputs
Mar 16, 2023	Microsoft integrates GPT-4 into its Office 365 suite, boosting productivity capabilities
Mar 21, 2023	Google unveils Bard, an AI chatbot based on the LaMDA family of LLMs
Apr 13, 2023	Amazon releases Bedrock, a comprehensive service offering multiple LLMs via API, including Amazon's Titan LLMs
May 2023	UAE's TII releases Falcon 40B model
July 18, 2023	Meta releases LLaMA 2, featuring open-source models with 7B, 13B, and 70B parameters
Sept 27, 2023	Mistral releases Mistral 7B which outperforms Meta's LLaMA 2 on several benchmarks

Source: Compiled by casewriters from McKinsey & Company, "What's the future of generative AI? An early view in 15 charts," August 15, 2023, <https://www.mckinsey.com/featured-insights/mckinsey-explainers/whats-the-future-of-generative-ai-an-early-view-in-15-charts>, accessed December 2023; Benj Edwards, "Meta launches Llama 2, a source-available AI model that allows commercial applications," Ars Technica, July 18, 2023, <https://arstechnica.com/information-technology/2023/07/meta-launches-llama-2-an-open-source-ai-model-that-allows-commercial-applications/>, accessed June 2024; Mistral AI, "Mistral 7B," September 27, 2023, <https://mistral.ai/news/announcing-mistral-7b/>, accessed June 2024; Lisa Barrington, "Abu Dhabi makes its Falcon 40B model open source," Reuters, May 25, 2023, <https://www.reuters.com/technology/abu-dhabi-makes-its-falcon-40b-ai-model-open-source-2023-05-25/>, accessed June 2024.

Exhibit 2 Other Members of Founding Team**Tom Brown**

Before founding Anthropic, Brown was the lead engineer for AI language model GPT-3 at OpenAI, and oversaw its development and deployment. He was also a member of the technical staff at Google Brain in the adversarial machine learning group, focusing on designing attacks and defenses for machine learning systems. Tom Brown graduated from Massachusetts Institute of Technology (MIT), where he earned a Master of Engineering in Computer Science and Brain and Cognitive Sciences.

Jack Clark

Clark served as the head of policy and communications at Anthropic. He was also a co-chair of the AI Index, an expert member of the Global Partnership on Artificial Intelligence, and a non-resident research fellow at the Center for Security and Emerging Technology. Before joining Anthropic, he was the Director of Policy at OpenAI, where he advocated for responsible and beneficial AI development. His background included a career in journalism, where he reported on technology and AI for various publications.

Jared Kaplan

Kaplan contributed to developing general AI systems and large language models. He previously held a position as a professor of physics at Johns Hopkins University, where his research spanned theoretical physics and quantum field theory. Before joining Anthropic, he was a member of the technical staff at OpenAI, contributing to several projects such as GPT-3 and DALL-E.

Benjamin Mann

Mann was a member of the technical staff at Anthropic. He previously worked as a member of the technical staff at OpenAI on infrastructure, efficiency, and safety for GPT-3. Mann had experience working at Google, the Machine Intelligence Research Institute, and at startups focusing on AI and automation. He studied computer science at Columbia University.

Sam McCandlish

Before joining Anthropic, McCandlish worked on the AI safety team at OpenAI, focusing on the science and scaling of machine learning, particularly scaling laws for neural language models, laying the groundwork for GPT-3. He was also a researcher at the Center for the Study of Existential Risk at the University of Cambridge, focusing on the long-term impacts and risks of AI. McCandlish graduated from Stanford University with a PhD. in theoretical physics.

Source: Compiled by casewriters from The Org, <https://theorg.com/org/anthropic/org-chart/tom-brown>, accessed December 2023; Salesforce Ventures, "In Conversation with Anthropic Co-founder Tom Brown," December 19, 2023, <https://salesforceventures.com/perspectives/in-conversation-with-anthropic-co-founder-tom-brown/>, accessed January 2024; Center for Security and Emerging Technology, <https://cset.georgetown.edu/staff/jack-clark/>, accessed January 2024; Stanford University Human-Centered Artificial Intelligence, <https://hai.stanford.edu/people/jack-clark>, accessed March 2024; John Hopkins University, <https://sites.krieger.jhu.edu/jared-kaplan/>, accessed March 2024; The AI Conference, <https://aiconference.com/speakers/ben-mann/>, accessed March 2024; The Org, <https://theorg.com/org/anthropic/org-chart/sam-mccandlish>, accessed March 2024.

Exhibit 3 Anthropic – Core Values**01****Here for the mission**

Anthropic exists for our mission: to ensure transformative AI helps people and society flourish. Progress this decade may be rapid, and we expect increasingly capable systems to pose novel challenges. We pursue our mission by building frontier systems, studying their behaviors, working to responsibly deploy them, and regularly sharing our safety insights. We collaborate with other projects and stakeholders seeking a similar outcome.

04**Do the simple thing that works**

We celebrate trying the simple thing before the clever, novel thing. We embrace pragmatism - sensible, practical approaches that acknowledge tradeoffs. We love empiricism - finding out what actually works by trying it - and apply this to our research, our engineering and our collaboration. We aim to be open about what we understand and what we don't.

02**Unusually high trust**

Our company is an unusually high trust environment: we assume good faith, disagree kindly, and prioritize honesty. We expect emotional maturity and intellectual openness. At its best, our trust enables us to make better decisions as an organization than any one of us could as individuals.

03**One big team**

Collaboration is central to our work, culture, and value proposition. While we have many teams at Anthropic, we feel the broader sense in which we are all on the same team working together towards the mission. Leadership sets the strategy, with broad input from everyone, and trusts each piece of the organization to pursue these goals in their unique style. Individuals commonly contribute to work across many different areas.

Source: Anthropic Website, "Our Values," <https://www.anthropic.com/company>, accessed December 2023.

Exhibit 4 Anthropic's AI Models

Anthropic's legacy models were Claude 2.0 (July 2023), Claude Instant 1.2 for businesses (August 2023), and Claude 2.1 (November 2023). The Claude 3 family of models: Haiku, Sonnet, and Opus (with Opus being the most powerful) were released in March 2024. These models demonstrated improved capabilities in analysis and forecasting, outperforming competitors like GPT-4 and Google's Gemini 1.0 Ultra in mathematical problem solving, computer coding, general knowledge and other areas. A significant advancement was Claude 3's ability to process both text and images, allowing analysis of diverse visual content, including photos, charts, graphs, and technical diagrams. Claude 3 could process multiple images (up to 20) in a single request, enabling comparative analyses (with some limitations). It did not generate artwork, and its models were strictly image-analyzing (for copyright and bias-related reasons). Claude 3 supported languages beyond English and was proficient at following multi-step instructions.

Opus was available to consumers at \$20 per month for a subscription while Sonnet was free. Claude's "context window" (the input text the model considered before generating additional text) was 200,000-tokens (or raw text)^e equivalent to about 150,000 words, with select customers getting up to a 1-million-token context window (or 700,000 words). Large-context models could better grasp the narrative flow of data and generated more contextually rich responses. Claude 3 could create expressive and engaging responses and was easier to prompt and steer compared to legacy models.

^e For example, the word "wonderful" would be split into the tokens "won," "der," and "ful."

Anthropic’s AI models were accessible through APIs (application programming interfaces), allowing developers to integrate them into their own applications. Anthropic focused on enterprise customers in sectors like healthcare, HR, financial services, sales, legal services, and education. Key partnerships included Zoom for AI product development, Quora’s integration of Claude in its Poe app, DuckDuckGo’s DuckAssist tool, and Notion AI’s writing assistant. With Claude 3, the company allowed businesses to build their own chatbots using the Opus and Sonnet technologies. The company listed GitLab, a GitHub competitor; TurboTax maker Intuit; and quantitative trading firm Jane Street as customers on its website.

Claude 3 Haiku is our fastest, most compact model for near-instant responsiveness. It answers simple queries and requests with unmatched speed. Users will be able to build seamless AI experiences that mimic human interactions.

Cost	\$0.25 \$1.25
<i>[Input \$/million tokens / Output \$/million tokens]</i>	
Context window	200K
Potential uses	• Customer interactions: quick and accurate support in live interactions, translations • Content moderation: catch risky behavior or customer requests • Cost-saving tasks: optimized logistics, inventory management, extract knowledge from unstructured data
Differentiator	Smarter, faster, and more affordable than other models in its intelligence category.

Claude 3 Sonnet strikes the ideal balance between intelligence and speed—particularly for enterprise workloads. It delivers strong performance at a lower cost compared to its peers, and is engineered for high endurance in large-scale AI deployments.

Cost	\$3 \$15
-------------	------------

*[Input \$/million tokens /
Output \$/million tokens]*

Context window	200K
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Potential uses	• Data processing: RAG or search & retrieval over vast amounts of knowledge • Sales: product recommendations, forecasting, targeted marketing • Time-saving tasks: code generation, quality control, parse text from images
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Differentiator	More affordable than other models with similar intelligence; better for scale.
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Claude 3 Opus is our most intelligent model, with best-in-market performance on highly complex tasks. It can navigate open-ended prompts and sight-unseen scenarios with remarkable fluency and human-like understanding. Opus shows us the outer limits of what's possible with generative AI.

Cost	\$15 \$75
-------------	-------------

*[Input \$/million tokens /
Output \$/million tokens]*

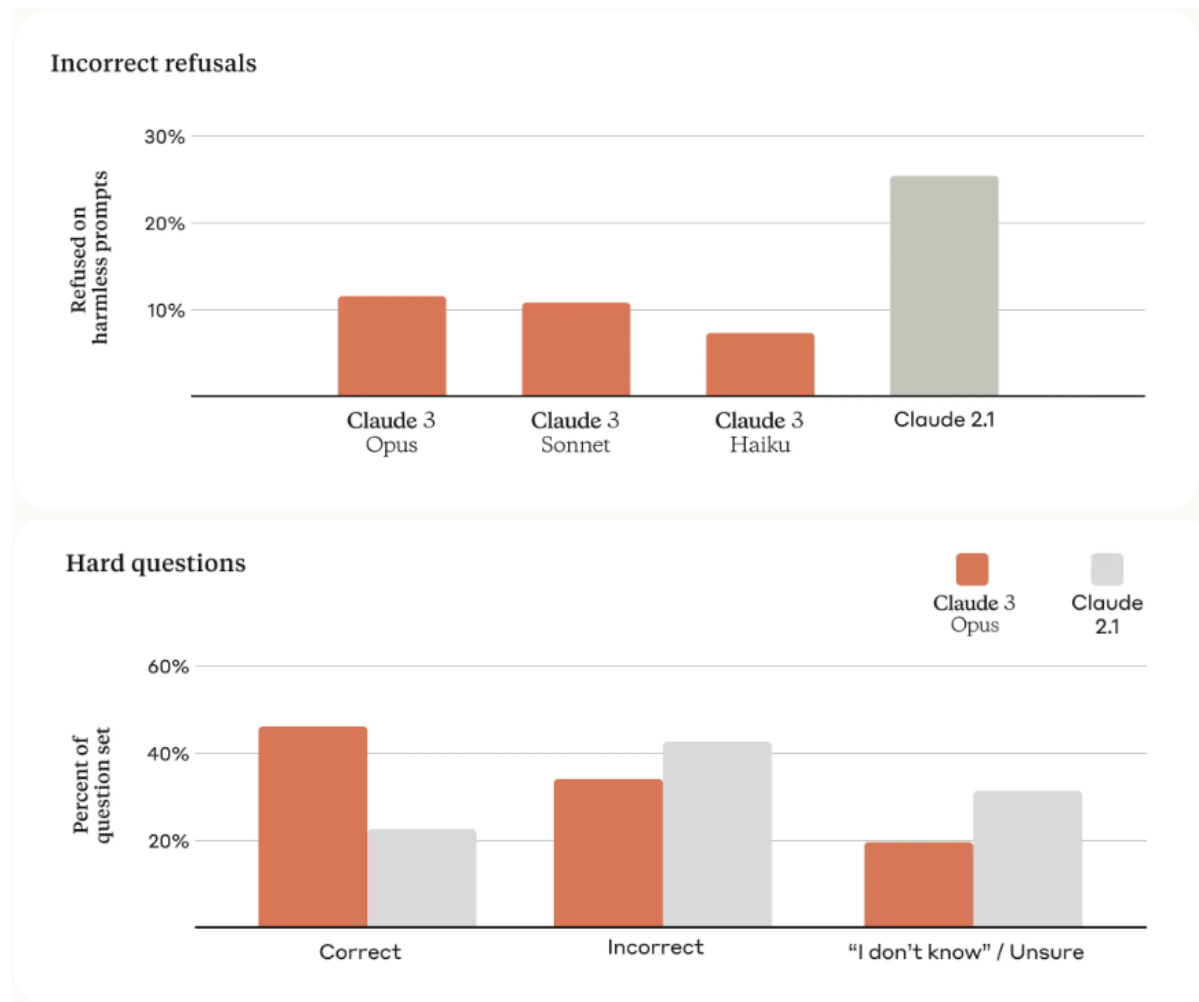
Context window	200K*
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Potential uses	• Task automation: plan and execute complex actions across APIs and databases, interactive coding • R&D: research review, brainstorming and hypothesis generation, drug discovery • Strategy: advanced analysis of charts & graphs, financials and market trends, forecasting
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Differentiator	Higher intelligence than any other model available.
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Claude 3 shared some inherent challenges with other GenAI models such as biases and hallucinations⁹⁴ (which meant the models could generate factually incorrect or irrelevant information and lacked the ability to check their output against a search of the web). The Claude 3 models had a more nuanced understanding of requests, recognized real harm, and refused to answer harmless prompts much less often compared to previous generations, indicating improved contextual understanding. Compared to Claude 2.1, Opus demonstrated a two-fold improvement in accuracy (i.e.,

correct answers) on challenging open-ended questions while also exhibiting reduced levels of incorrect answers. Soon, Claude would be able to cite sources allowing users to verify answers.⁹⁵ Overall, guardrails with Claude were meant to be stronger and more dependable than competitors. Although Anthropic's business volume was lower than OpenAI's, Anthropic excelled in predicting behaviors and uses of AI models (such as how bad actors may harness them in cyberattacks).⁹⁶



The upgraded Claude 3.5 Sonnet, released in October 2024, improved performance across the board, excelling further in coding. Claude 3.5 Haiku matched the capabilities of the previous flagship, Claude 3 Opus, while maintaining the speed of the earlier Haiku generation.

Source: Compiled by casewriters from Anthropic Website, "Introducing the next generation of Claude," March 4, 2024, <https://www.anthropic.com/news/claude-3-family>, accessed March 2024; Cade Metz, "A.I. Start-Up Anthropic Challenges OpenAI and Google With New Chatbot," *The New York Times*, March 4, 2024; Kyle Wiggers, "Anthropic claims its new AI chatbot models beat OpenAI's GPT-4," *TechCrunch*, March 4, 2024, <https://techcrunch.com/2024/03/04/anthropic-claims-its-new-models-beat-gpt-4/?guccounter=1>, accessed March 2024; Anthropic website, "Zoom partnership and investment in Anthropic," May 16, 2023, <https://www.anthropic.com/index/zoom-partnership-and-investment>, accessed October 2023; Sarah Perez, "Quora opens its new AI chatbot app Poe to the general public," *TechCrunch*, February 6, 2023, <https://techcrunch.com/2023/02/06/quora-opens-its-new-ai-chatbot-app-poe-to-the-general-public/>, accessed September 2023; Kyle Wiggers, Devin Coldeway, and Manish Singh. "Anthropic's \$5B, 4-year plan to take on

OpenAI,” *TechCrunch*, April 6, 2023, <https://techcrunch.com/2023/04/06/anthropics-5b-4-year-plan-to-take-on-openai/>, accessed September 2023; Kyle Wiggers, “Anthropic claims its new AI chatbot models beat OpenAI’s GPT-4,” *TechCrunch*, March 4, 2024, <https://techcrunch.com/2024/03/04/anthropic-claims-its-new-models-beat-gpt-4/?guccounter=1>, accessed March 2024; Maria Heeter, Anissa Gardizy, and Natasha Mascarenhas, “Anthropic to Raise \$750 Million in Menlo-Ventures Led Deal,” *The Information*, December 20, 2023, <https://www.theinformation.com/articles/anthropic-to-raise-750-million-in-menlo-ventures-led-deal?rc=kimdxv>, accessed December 2023.

Exhibit 5 Regulatory Framework

Following the debut of ChatGPT, rapid advancements in AI raised alarms in the tech community. The Center for AI Safety, a San Francisco-based non-profit released a joint statement from CEOs and scientists from tech companies, “Mitigating the risk of extinction from AI should be a global priority alongside other societal-scale risks such as pandemics and nuclear war.” The warning had the effect of making the technology sound groundbreaking. The *Atlantic* responded, “The CEOs, like demigods, are wielding a technology as transformative as fire, electricity, nuclear fission, or a pandemic-inducing virus. You’d be a fool not to invest.” Altman testified in the U.S. Congress and advocated for AI regulation through licensing.

By July 2023, questions arose about the ability of tech companies to self-regulate. Following a meeting with President Biden, seven leading AI businesses publicly committed to safety and transparency measures. Four of these companies, including Anthropic, Google, Microsoft, and OpenAI, subsequently formed a coalition to investigate more potent AI forms and establish best practices for their control. The White House commitments were voluntary and not enforceable, but signaled that the companies and the government were working on AI safety.

Despite these commitments, there were concerns about the tech industry’s record in mitigating harms, as with social media. This led to calls for a regulatory framework to ensure these pledges were upheld without hindering innovation. The firms had promised pre-release security testing and information sharing on risk management, among other measures. The EU, in its regulatory approach, had already proposed an AI Act, while the U.S. Congress seemed less prepared to legislate on AI. The consensus was that effective AI regulation would necessitate close collaboration with the tech sector, incorporating insights from both internal and external AI experts. The *Financial Times* commented, “Ultimately, legislation is needed but this runs into issues of proportionality, and lawmakers’ ability to draw up safeguards on a sophisticated technology evolving at warp speed.” Technologist Tristan Harris pointed to the complexity gap, “The issue here is that the complexity and speed and power of technology is scaling way faster than the level of complexity of our governance. That’s the meta issue [...] that we have to solve.”

In October 2023, the US government issued an Executive Order on AI, to address a range of AI-related risks, including privacy, fairness, bias, and catastrophic risks. It recognized AI’s potential benefits, directing government departments to identify opportunities to harness the technology to improve government services and functions by creating Chief AI Officers and promoting AI innovation in their agencies. The next month, the EU passed a new set of comprehensive AI regulations that placed strong restrictions of high-risk AI systems. These restrictions included requirements for transparency, safety, accountability, data governance, documentation and human oversight. The act also prohibited certain practices such as systems that manipulate human behavior or systems that allow “social scoring” by governments.

Source: Compiled by casewriters from Cristina Criddle, “Tech chiefs say AI poses similar threat to nuclear war,” *The Financial Times*, May 31, 2023; Lora Kelley, “AI Companies Are Trying to Have It Both Ways,” *The Atlantic*, July 24, 2023; The

White House. "Fact Sheet: Biden-Harris Administration Secures Voluntary Commitments from Leading Artificial Intelligence Companies to Manage the Risks Posed by AI," July 21, 2023. <https://www.whitehouse.gov/briefing-room/statements-releases/2023/07/21/fact-sheet-biden-harris-administration-secures-voluntary-commitments-from-leading-artificial-intelligence-companies-to-manage-the-risks-posed-by-ai/>, accessed October 2023; The Editorial Board, "Can tech companies police themselves on AI," *The Financial Times*, July 29, 2023; Tristan Harris and Aza Raskin, "Inside the First AI Insight Forum in Washington," *Your Undivided Attention* Podcast, Centre for Humane Technology, September 18, 2023.

Exhibit 6 Anthropic's Research Principles

(1) AI as a Systematic Science

Inspired by the universality of scaling in statistical physics, we develop scaling laws to help us do systematic, empirically-driven research. We search for simple relations among data, compute, parameters, and performance of large-scale networks. Then we leverage these relations to train networks more efficiently and predictably, and to evaluate our own progress. We're also investigating what scaling laws for the safety of AI systems might look like, and this will inform our future research.

(2) Safety and Scaling

At Anthropic we believe safety research is most useful when performed on highly capable models. Every year, we see larger neural networks which perform better than those that came before. These larger networks also bring new safety challenges. We study and engage with the safety issues of large models so that we can find ways to make them more reliable, share what we learn, and improve safe deployment outcomes across the field. Our immediate focus is prototyping systems that pair these safety techniques with tools for analyzing text and code.

(3) Tools and Measurements

We believe critically evaluating the potential societal impacts of our work is a key pillar of research. Our approach centers on building tools and measurements to evaluate and understand the capabilities, limitations, and potential for societal impact of our AI systems. A good way to understand our research direction here is to read about some of the work we've led or collaborated on in this space.

(4) Focused, Collaborative Research Efforts

We highly value collaboration on projects, and aim for a mixture of top-down and bottom-up research planning. We always aim to ensure we have a clear, focused research agenda, but we put a lot of emphasis on including everyone — researchers, engineers, societal impact experts and policy analysts — in determining that direction. We look to collaborate with other labs and researchers, as we believe the best research into characterizing these systems will come from a broad community of researchers working together.

Source: Anthropic Website, "Make safe AI systems, Deploy them reliably," <https://www.anthropic.com/research>, accessed January 2024.

Exhibit 7 RLHF (Reinforcement Learning from Human Feedback)

Paul Christiano, a former scientist at OpenAI and current member of Anthropic's LTBT, was instrumental in developing the RLHF concept. The foundation of RLHF was reinforcement learning, a foundational AI technique since the 1980s. This was centered around creating an agent (a software program or a robot), and guiding its actions through a system of rewards. For instance, if the goal were to teach a robot to navigate a maze, rewards might be given each time the robot moved closer to the exit without hitting a wall.

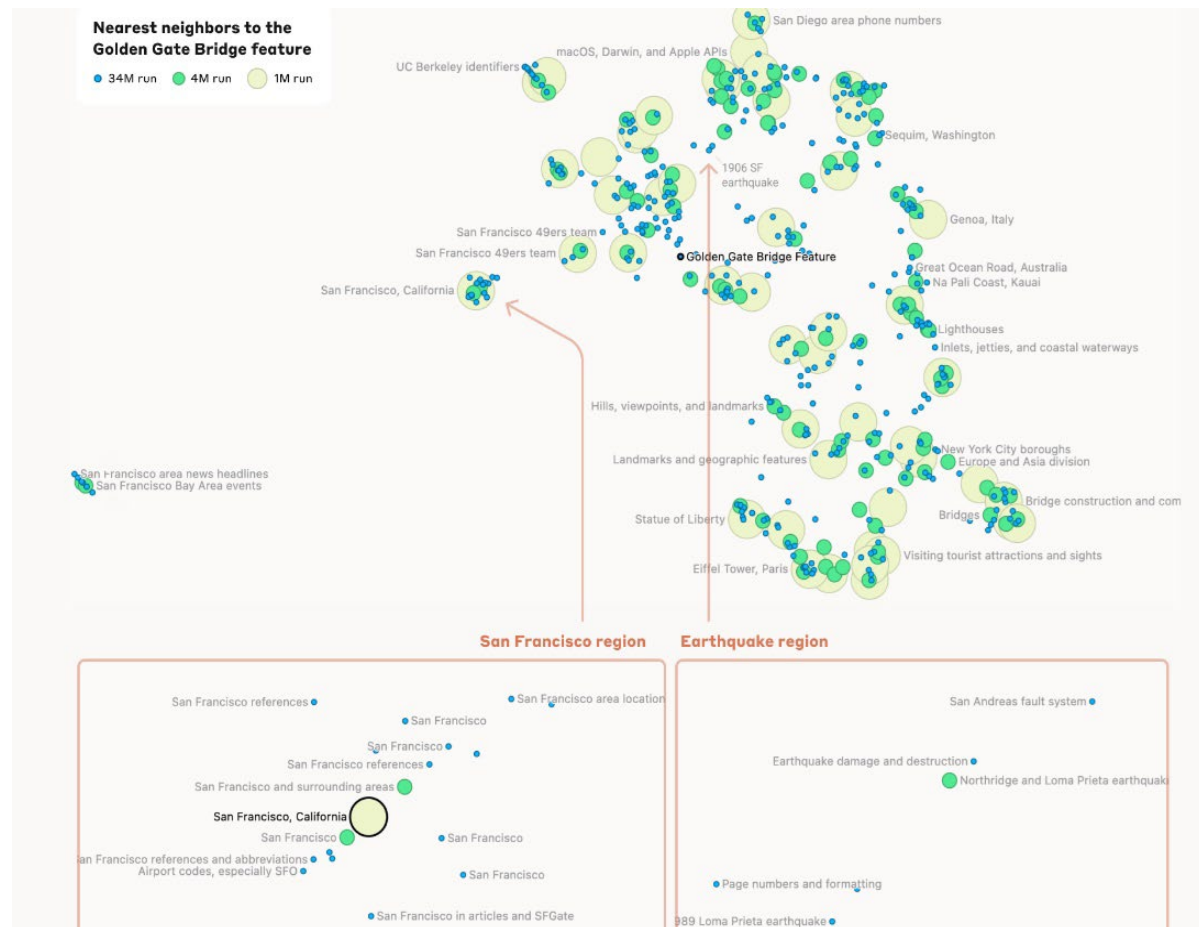
In gaming, the reward system was relatively straightforward. For example, a chess-playing AI might be rewarded for each game it won, mirroring the principles behind DeepMind's AlphaZero chess AI. However, language models presented more complex reward parameters. Users expected a chatbot like Claude not only to provide accurate responses but also to perform diverse tasks, such as solving mathematical problems, interpreting music, and exhibiting human-like creativity and understanding, without bias – all while remaining under human control. Expressing these intricate expectations for an AI system was a daunting task, leading to the development of RLHF, which sought guidance directly from humans.

Introduced in 2017 by Dario and his team, the RLHF method wasn't applied to language models until much later. Initially, models were trained on extensive text data without specific instructions, aiming for content prediction. The distinctiveness of RLHF was evident in the subsequent phase, where human evaluators (mainly from the global south) were employed to review the model's outputs. For controversial queries, such as opinions on political figures, the model could produce a range of answers. Contractors evaluated these outputs, determining the most appropriate response based on company guidelines. This feedback loop, involving thousands of evaluations, refined the model's responses.

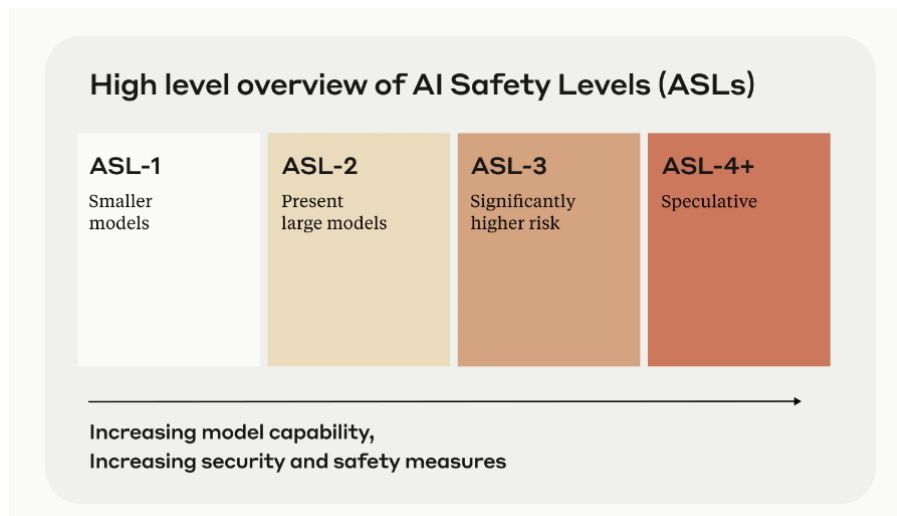
Although RLHF was able to guide the model towards desired outputs (whether politically neutral or possibly biased), it wasn't without its challenges. A significant issue was the lack of transparency. When questioned about the rationale behind a specific answer, the explanations were often vague. The model's response were based on a culmination of numerous contractors' evaluations and the model's statistical interpretation of that feedback, making the process opaque and hard to explain. Furthermore, modifying the model's behavior posed difficulties. If the model's outputs were unsatisfactory, communicating the necessary changes to a vast contractor base and ensuring consistent understanding was daunting. Anthropic's Constitutional AI was developed to address some of these challenges by allowing developers to specify values explicitly, separating technical compliance from value judgments, thus aiming for a more transparent and reliable alignment process.

Source: Compiled by casewriters from Dylan Matthews, "The \$1 billion gamble to ensure AI doesn't destroy humanity," September 25, 2023, *Vox*, <https://www.vox.com/future-perfect/23794855/anthropic-ai-openai-claude-2>, accessed October 2023; Will Henshall, "Paul Christiano: The 100 Most Influential People in AI," *Time*, September 7, 2023.

Exhibit 8 The Golden Gate Bridge Feature



Source: Adly Templeton, Tom Conerly, et. al, "Scaling Monosemanticity: Extracting Interpretable Features from Claude 3 Sonnet," Anthropic, May 21, 2024, <https://transformer-circuits.pub/2024/scaling-monosemanticity/index.html>, accessed July 2024.

Exhibit 9 AI Safety Levels

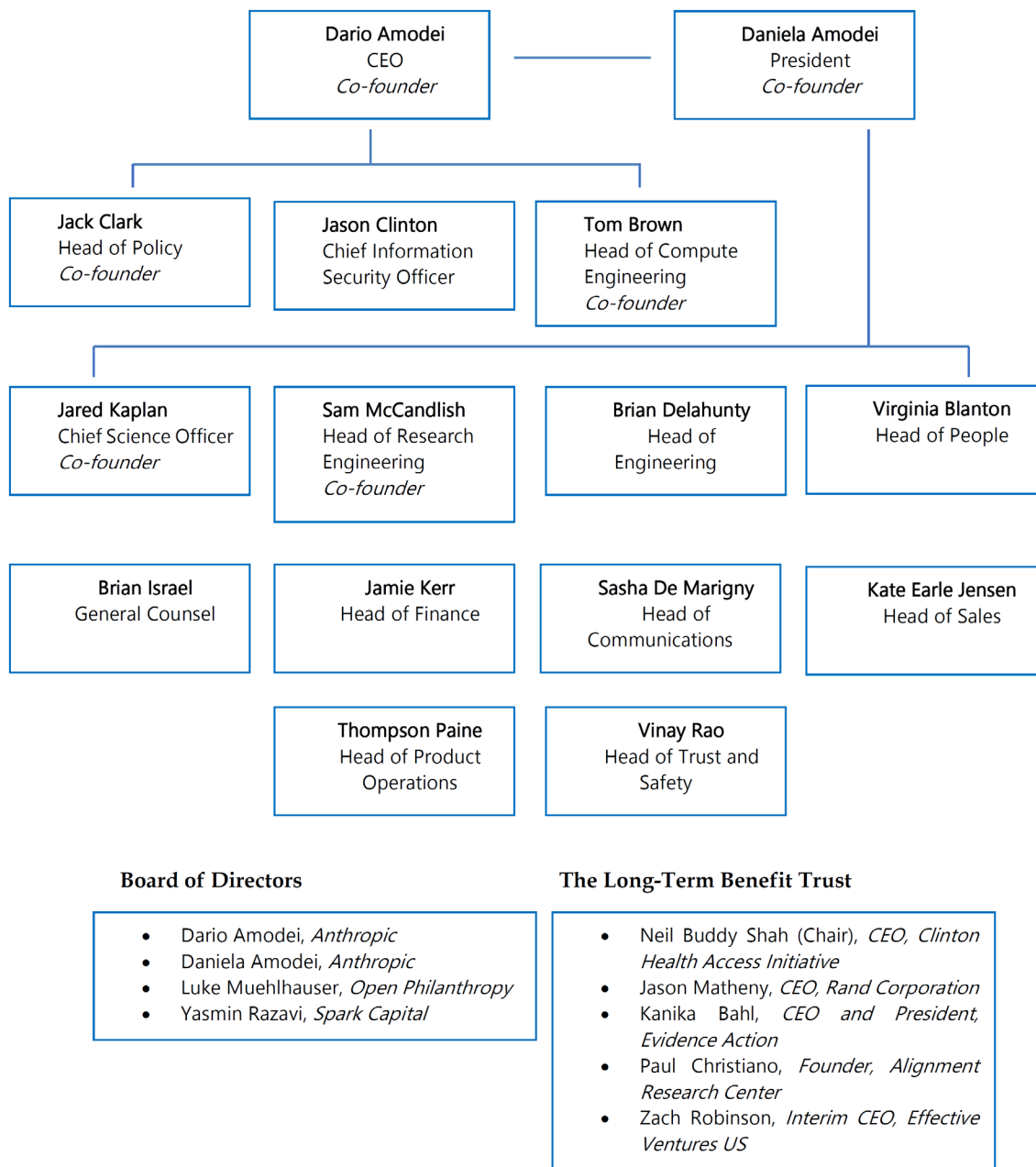
ASL-1: systems that posed no significant catastrophic risk, such as a 2018 LLM or an AI system designed solely for playing chess.

ASL-2: systems that exhibited early indications of hazardous capabilities, like providing guidance on bioweapon creation. However, this information wasn't considered useful due to its lack of reliability or because it wasn't offering anything beyond what a search engine might provide. Current LLMs, including Claude, seemed to fall under ASL-2.

ASL-3: systems that significantly heightened the risk of catastrophic misuse compared to non-AI benchmarks, such as search engines or textbooks, or those that demonstrated basic autonomous capabilities.

ASL-4 and higher levels (ASL-5+): not defined as they were considered too advanced compared to existing systems. However, they were expected to encompass major increases in potential for catastrophic misuse and autonomy.

Source: Anthropic Website, "Anthropic's Responsible Scaling Policy," September 19, 2023, <https://www.anthropic.com/index/anthropics-responsible-scaling-policy>, accessed December 2023.

Exhibit 10 Organizational Structure (as of December 2023)

Source: Adapted from Akash Pasricha, "Anthropic Org Chart," *The Information*, December 12, 2023, <https://www.theinformation.com/org-charts/anthropic?rc=kimdxv>, accessed January 2024.

Exhibit 11 Monetization Strategy and Fundraising

Many of Anthropic's early hires and investors were associated with effective altruism (EA), a utilitarian-inspired movement rooted in Silicon Valley which believed in using data and logic to maximize positive impact and further "the good." Effective altruists were early advocates for addressing existential risks posed by AI, leading many supporters to pursue careers in AI safety. Although Anthropic did not identify as an EA company, EA pervaded its ethos.⁹⁷ Dario commented, "I'm sympathetic to a number of the [EA] ideas. I want Anthropic to be about solving the problems that it's solving, not about some particular movement."⁹⁸

Early investors in both Series A (\$124M raised) and Series B (\$580M raised) were strongly aligned with the safety goals and tradeoffs inherent in Anthropic's mission. Daniela explained, "If the only thing that [investors] care about is return on investment, we just might not be the right company for them to invest in. And that's something that we are very open about when we are fundraising." Dario described the mixed reactions of investors: "Some of them are just like, whatever, you run your company how you want. Some of them are like, oh my god, this body of random people could move Anthropic in a direction that's contrary to shareholder value. And then it gets into a conversation of, well, what are the kinds of things that we might do that would be contrary to the interests of traditional investors."

In April 2023, the company's pitch deck (obtained by *TechCrunch*) was geared towards a Series C raise: "We've developed a strategy for go-to-market and initial product specialization that fits with our core expertise, brand and where we see adoption occurring over the next 12 months." Anthropic planned to build a frontier model that was 10 times more capable than today's most powerful AI. The deck described the frontier model as a "next-gen algorithm for AI self-teaching," referring to constitutional AI. The company planned to raise \$5 billion over the next two years for this chatbot, a successor to Claude, which was expected to cost \$1 billion to develop over 18 months. The frontier model would require on the order of 10^{25} FLOPs (or floating-point operations) — several orders of magnitude larger than even the biggest models at the time. This model could "begin to automate large portions of the economy. We believe that companies that train the best 2025/26 models will be too far ahead for anyone to catch up in subsequent cycles."

Anthropic raised its \$450 million Series C round in May 2023, bringing total funding to \$1.5 billion at a valuation of \$5 billion. Series C investors included financial entities such as Spark Capital and strategic partners like Google, which held a 10% stake with its \$300 million investment. Google Cloud would be Anthropic's "preferred cloud provider," and the two entities would jointly create AI computing systems. A few months later in September, Amazon announced an investment of \$4 billion into the company, signaling a deep interest in joining the AI race. Amazon would gain minority ownership of the company and incorporate Anthropic's technology into various products across its businesses, including the Amazon Bedrock service for building AI applications.⁹⁹ Anthropic would use Amazon's custom chips to build, train, and deploy its future AI foundation models, and it committed that Amazon Web Services (AWS) would be its primary cloud provider.

Following Amazon and Google's investments, Anthropic signed cloud computing agreements, which committed it to spend \$3 billion on the Google Cloud Platform over four years and \$4 billion on AWS over five years. This implied that Anthropic would spend well over \$1 billion a year on cloud services. In February 2024, Menlo Ventures closed a deal to invest \$750 million in Anthropic. Menlo created a "special purpose vehicle" to consolidate the smaller investors interested in Anthropic. In the year from 2023 to 2024, Anthropic's valuation tripled to \$15 to \$20 billion – 3x-4x the valuation of the

previous round in the spring. Anthropic was expected to generate revenues of \$200 million in 2023 and between \$500 and \$850 million in 2024.

By the fall of 2024, Anthropic began discussions with investors about a new funding round that could value the company between \$30-\$40 billion, approximately 50 times Anthropic's projected annualized revenue of \$800 million for the year. In November, Amazon invested an additional \$4 billion in Anthropic, reflecting the tech giant's urgency to compete in the rapidly evolving AI sector. The investment brought Anthropic's total venture capital funding to \$13.7 billion.

Source: Compiled by casewriters from Dylan Matthews, "The \$1 billion gamble to ensure AI doesn't destroy humanity," September 25, 2023, *Vox*, <https://www.vox.com/future-perfect/23794855/anthropic-ai-openai-claude-2>, accessed October 2023; Dwarkesh Patel, "Dario Amodei - Scaling, Alignment & AI Progress," Dwarkesh Podcast, August 8, 2023, <https://www.dwarkeshpatel.com/p/dario-amodei#details>, accessed September 2023; Kyle Wiggers, Devin Coldeway, and Manish Singh, "Anthropic's \$5B, 4-year plan to take on OpenAI," *TechCrunch*, April 6, 2023, <https://techcrunch.com/2023/04/06/anthropics-5b-4-year-plan-to-take-on-openai/>, accessed September 2023; James Vincent, "Google invested \$300 million in AI firm founded by former OpenAI researchers," *The Verge*, February 3, 2023, <https://www.theverge.com/2023/2/3/23584540/google-anthropic-investment-300-million-openai-chatgpt-rival-claude>, accessed October 2023; Jess Weatherbed, "Amazon will invest up to \$4 billion into OpenAI rival Anthropic," *The Verge*, September 25, 2023, <https://www.theverge.com/2023/9/25/23888841/amazon-4-billion-investment-anthropic-claude-ai-openai-microsoft>, accessed October 2023; Maria Heeter, Anissa Gardizy, and Natasha Mascarenhas, "Anthropic to Raise \$750 Million in Menlo-Ventures Led Deal," *The Information*, December 20, 2023, <https://www.theinformation.com/articles/anthropic-to-raise-750-million-in-menlo-ventures-led-deal?rc=kimdxv>, accessed December 2023; Erin Griffith and Cade Metz, "Inside the Funding Frenzy at Anthropic, One of A.I.'s Hottest Start-ups," *The New York Times*, February 20, 2024, <https://www.nytimes.com/2024/02/20/technology/anthropic-funding-ai.html>, accessed March 2024; Anissa Gardizy and Kate Clark, "Anthropic Has Floated \$40 Billion Valuation in Funding Talks," *The Information*, September 23, 2024, <https://www.theinformation.com/articles/openai-rival-anthropic-has-floated-40-billion-valuation-in-early-talks-about-new-funding?rc=kimdxv>, accessed November 2024; Adam Satariano, "Amazon Invests \$4 Billion in Anthropic, Deepening its A.I. Ties," *The New York Times*, November 22, 2024, <https://www.nytimes.com/2024/11/22/technology/amazon-anthropic-ai.html>, accessed November 2024; Kyle Wiggers, "Anthropic Raises another \$4B from Amazon, makes AWS its 'primary' training partner," *TechCrunch*, November 22, 2024, <https://techcrunch.com/2024/11/22/anthropic-raises-an-additional-4b-from-amazon-makes-aws-its-primary-cloud-partner/>, accessed November 2024.

Exhibit 12 Varying Governance Structures

OpenAI OpenAI, initially a non-profit research organization founded in 2015, was restructured into a hybrid entity in 2019, comprising OpenAI, Inc. (a non-profit) and OpenAI LP (a capped-profit subsidiary). The board was responsible for safe development of AI and oversaw the non-profit that set the vision and mission. The subsidiary focused on capital raising and R&D, with profits capped at 100 times the initial investment, and any surplus directed to the non-profit. The board was obligated to OpenAI's mission of "ensuring the creation and adoption of safe and beneficial AGI" rather than financial incentives. The CEO Sam Altman, owning no shares, envisioned this broader goal. The change in OpenAI's structure drew criticism from initial supporters like Elon Musk, who compare this to turning a rainforest-conservation organization into a lumber outfit. In 2019, Microsoft invested \$1 billion in OpenAI, becoming its exclusive cloud provider. The partnership further solidified in 2023 with a \$10 billion investment. This commercialization strategy sparked controversy within the AI community and among the company's leadership. The unusual governance structure was aimed to balance morality and financial strength, but led to internal conflict and the firing (and eventual reinstatement) of Altman as CEO in November 2023.

DeepMind and Google Founded in 2010 and acquired by Google in 2014, DeepMind was a UK-based AI research lab primarily serving Alphabet and its subsidiaries, focusing on high-profile research projects without directly selling products to consumers. As part of Alphabet, DeepMind faced scrutiny over the profitability of its technology, influencing its research directions. With a mission "to

advance science and benefit humanity,” DeepMind had a separate unit called Deep Mind Ethics and Society, established in 2017 to explore the ethical and social implications of AI. It established an ethics board – which only met once. In April 2023, in response to the AI advancements like ChatGPT, Google merged DeepMind with Google Brain to form “Google DeepMind.” This merger aimed to combine DeepMind’s research prowess, known for the AlphaGo project, with Google Brain’s incremental product enhancements and transformer architecture expertise. The merger sought to revitalize Google’s AI innovation and adopt a more product-focused approach, uniting the previously rival teams to create next-generation AI breakthroughs and products. A blog post explained, “Through Google DeepMind, we are bringing together our world-class talent in AI with the computing power, infrastructure and resources to create the next generation of AI breakthroughs and products across Google and Alphabet, and to do this in a bold and responsible way.”

Meta Meta’s open-source model was a term referring to the company’s practice of releasing some of its AI research and development projects to the public domain, allowing anyone to access, use, modify, and improve them. The model aimed to foster collaboration and innovations in the AI field and contribute to social good. For example, LLaMA was a large language model that used AI to generate text, images, and code from natural language queries. An open-source version of LLaMA called LLaMA 2 was released for public use in July 2023. Meta released a statement on its open approach: “We support an open innovation approach to AI. Responsible and open innovation gives us all a stake in the AI development process, bringing visibility, scrutiny and trust to these technologies. Opening today’s Llama models will let everyone benefit from this technology.”

Source: Compiled by casewriters from Shikhar Ghosh and Shweta Bagai, “OpenAI: Idealism Meets Capitalism,” Harvard Business School Case 824-134, January 2024; Ben Dickson, “AI lab DeepMind becomes profitable and bolsters relationship with Google,” *VentureBeat*, October 10, 2021, <https://venturebeat.com/ai/ai-lab-deepmind-becomes-profitable-and-bolsters-relationship-with-google/>, accessed October 2023; <https://ai.meta.com/llama/>; Ross Andersen, “Does Sam Altman Know What He’s Creating?” *The Atlantic*, July 24, 2023.

Exhibit 13 Select LLMs Developers

Developers of LLMs offered models and APIs that enabled enterprises to build natural language processing (NLP) applications for content creation, summarization, classification, chat, and sentiment analysis. Enterprises could fine-tune and customize these large-scale language models for their specific use cases. A few of the key models are described below.

OpenAI OpenAI was founded as a non-profit research organization in 2015 by a group of tech entrepreneurs (Elon Musk, Peter Thiel, Reid Hoffman, and others). Its goal was to advance AGI “in a way that is most likely to benefit humanity as a whole, unconstrained by a need to generate financial return.” In 2019, OpenAI restructured into two organizations: OpenAI Nonprofit and OpenAI LP. The nonprofit would remain a 501(c)(3) operating under its original charter, while the LP would be a “capped-profit” partnership overseen by the nonprofit. The structure was intended to attract new investors without compromising its mission: profits for investors in the LP would be capped at 10 percent of investment. In July 2019, Microsoft announced a \$1 billion investment in OpenAI and became the exclusive cloud provider, collaborating on a hardware and software program to incorporate AGI within Azure.

One year later, in June 2020, OpenAI released GPT-3, a large language model with 175 billion parameters. It was the first model to offer a glimpse of mainstream usability through a public-facing API, and marked a turning point in language-based AI. GPT-3 could generate realistic text and perform various tasks based on natural language prompts. OpenAI offered various generative AI products through its API service, such as Access GPT-3, DALL-E 2, and Codex. Microsoft announced in September that it would exclusively license OpenAI’s GPT-3 model. OpenAI launched ChatGPT, a chatbot interface built on GPT-3.5, in November 2022. The chatbot captured widespread attention and reached 100 million active users in two months. ChatGPT could converse in humanlike fashion and supported plug-ins to access external sites and services. A UBS study indicated that ChatGPT was the fastest growing consumer application in history. OpenAI spent over \$540 million in developing ChatGPT, reflecting the high costs in training LLMs, and pointed to the reasons for OpenAI’s dependence on Microsoft for computing power and cash.

In January 2023, OpenAI and Microsoft announced a “multiyear, multibillion dollar” extension to their partnership with a new investment from Microsoft. Terms of the deal were not disclosed, but Microsoft’s investment was reported to be \$10 billion. There were rumors that Microsoft would receive 75% of OpenAI’s profits until it secured its investment return, and a 49% stake in the company. Microsoft would become the exclusive cloud partner for OpenAI going forward and would provide unlimited computing power from its Azure cloud. Microsoft began deploying OpenAI’s models in its Bing search engine, and would roll out AI features in other products such as Office and Teams. Microsoft would continue to license OpenAI’s GPT-3 model exclusively, while OpenAI could still offer its API service to third-party developers.

OpenAI released GPT-4, a large language model with an estimated 1.5 trillion parameters in March 2023. It was the most advanced and versatile model that could generate realistic text and images on several topics and tasks. GPT-4 demonstrated improved conversational abilities, responsiveness to user steering, potential for image-based inputs, and safety precautions to prevent harmful advice or inappropriate content. GPT-4 also achieved human-level performance on various professional and academic benchmarks. GPT’s technical documentation did not include any details on the architecture (including model size), hardware, training compute, data set, or training method. OpenAI had signaled very clearly its shift towards a closed model, proprietary training methods and the non-disclosure of training data. OpenAI’s chief scientist claimed that open-sourcing AI was not wise. In November 2023,

Altman announced a series of new developments: larger context windows, more economical pricing, and the ability for users to create their own 'agents' within the GPT framework.

Microsoft Microsoft began its journey in natural language processing (NLP) in 1990 with the establishment of Microsoft Research, recruiting top computational linguists from IBM. The company pioneered grammar detection, spell check, and translation tools, cementing its position as an NLP leader. By 2014, Microsoft introduced Azure ML, one of the first cloud-based machine learning platforms. Through collaborations with global academic researchers, Microsoft's Turing program created LLMs like Turing-NLG (17 billion parameters) and Megatron-Turing NLG (530 billion parameters). In 2019, a \$1 billion investment in OpenAI made Microsoft its exclusive cloud provider. This partnership deepened in 2020 with an exclusive license to OpenAI's GPT-3 model and further solidified in 2023 with a \$10 billion investment. Microsoft integrated OpenAI's models into products like Bing, Office, and Teams.

Generative AI features enhanced Bing with conversational answers and chat capabilities. Microsoft 365 also benefited, introducing AI-assisted tools for writing, data analysis, and coding. The company's vision shifted from AI autonomy to AI collaboration, emphasizing human-centric design. While leveraging OpenAI's advancements, Microsoft maintained its in-house AI development. Bing's conversational AI often relied on Microsoft's efficient Turing language models instead of GPT-4, for more cost-effective operations. While Microsoft remained committed to its partnership with OpenAI (which exclusively powered Microsoft's first-party AI apps such as Copilot), Microsoft increased its development of small models and continued to broaden the range of models offered through its Azure cloud platform. It added Cohere's latest LLM to Azure, and forged a deal with French AI company Mistral. In March 2024, Microsoft appointed DeepMind co-founder Mustafa Suleyman as the head of a newly created consumer AI unit, and hired several employees of his Inflection AI startup. The new unit would combine consumer AI efforts such as Microsoft's Copilot chatbot and integrate AI into Bing and the Edge web browser.

Google Google had a long-standing history in AI research and development, both internally and through acquisitions. The company made notable strides in NLP, LLMs, and pre-training tools. The company's researchers devised the transformer architecture which paved the way for innovations like GPT. While Google boasted a decade of experience in AI and ML technologies, much of its work was conducted in-house. From 2009, Google acquired over 30 startups with expertise in AI and ML, with its AI-related acquisitions reaching \$3.8 billion in 2020. In 2011, Google Brain began as a project aimed at exploring neural networks and general-purpose AI technologies. Google acquired DeepMind in 2014 for \$500 million. DeepMind was an AI company that utilized games to train and test its AI systems. Notably, in 2016, its AlphaGo program defeated a world champion in *Go*, a challenging strategy game. DeepMind functioned as a separate entity under Alphabet, distinct from the divisions that encompassed Brain. However, this changed in April 2023 when DeepMind and Brain merged under Google Research.

In 2018, Google open-sourced BERT (Bidirectional Encoder Representations), an NLP pre-training technique. This method was widely adopted by subsequent LLMs and played a pivotal role in the rise of chatbots like GPT-3. In 2020, following OpenAI's announcement that GPT-3 would be exclusively licensed to Microsoft, Google intensified its efforts on LLMs. In April 2022, Google unveiled its 540 billion parameter Pathway Language Model (PaLM), which was trained using a system from Google Brain. By January 2023, reports suggested that Google was gearing up to launch a range of generative AI products. Subsequently, in March 2023, Google introduced its chatbot, Bard, with 137 billion parameters; Bard had mixed reviews. The company merged its two leading A.I. labs — Google Brain

and DeepMind — and announced that the combined lab was developing a new A.I. technology called Gemini.

Gemini was introduced in February 2024, and combined features of both a talking digital assistant and a conversational chatbot. Gemini responded to both voice and text inputs, enabling it to answer questions, compose poetry, generate images, draft emails, and analyze personal photos. Available to English speakers in over 150 countries, Gemini superseded both Bard and Google Assistant. Gemini was designed to assist with a range of activities including tutoring, coding assistance, and interview preparation. The app was also multimodal, capable of handling inputs like images and sounds. However, Google disabled its image generator after Gemini created images showing people of color in German military uniforms from World War II. The Gemini app had two versions: a free limited version and a more advanced subscription-based Gemini Advanced (underpinned by Google's Ultra 1.0 language model).

Meta Meta AI, previously known as Facebook AI Research (FAIR), was the AI division of Meta, established in 2013. It delved into areas like NLP, computer vision, and generative adversarial networks (GANs). Over the years, Meta AI introduced models like fastText, Blenderbot, OPT-175B, LLaMA, and CICERO, and maintained the widely-used machine learning framework, PyTorch. In February 2023, Meta AI launched LLaMA, a series of language models with sizes ranging from 7 billion to 65 billion parameters. The 13 billion-parameter LLaMA outstripped GPT-3 in many benchmarks, while LLaMA-65B rivaled the world's top LLMs. After releasing LLaMA under a non-commercial GPL 3.0 license for the research community, unauthorized copies surfaced, effectively making LLaMA open-source. Meta AI also disclosed the model's construction and training details, offering a rare insight into advanced LLM development for researchers outside major labs. In April 2024, Meta released Llama 3, featuring models with 8B and 70B parameters that could support a broad range of use cases. Llama 3, along with other models like Llama 2 and Code Llama, were available on various platforms, reflecting Meta's strategy to democratize access.

Baidu Baidu, the dominant internet search company in China, released its chatbot, Ernie Bot, accessible via app stores and Baidu's website. The bot offered functionalities similar to its competitor, ChatGPT. Users could interact with Ernie Bot for a range of tasks, from asking questions to seeking assistance in writing market analyses, crafting marketing slogans, and summarizing documents. A Chinese phone number was mandatory for registration and login. Within the first 19 hours post-launch, Ernie Bot's user base skyrocketed, surpassing one million. Baidu further expanded Ernie Bot's ecosystem by introducing a dedicated plug-in market. An initial demonstration by Baidu, which utilized prerecorded videos, drew criticism and left investors underwhelmed. Before securing the government's approval for a commercial rollout, Baidu, along with several other AI enterprises, underwent rigorous security assessments to ensure adherence to China's stringent generative AI guidelines. These guidelines mandated companies to align with the "core values of socialism" and utilize only government-approved sources for training foundation models. The company hinted at plans to introduce a series of AI-centric applications designed to display generative AI's core capabilities. Recent data indicated that the bot had over 200 million users, cementing its position as China's most popular ChatGPT-like chatbot.

TII Falcon 180B was a language model introduced by Technology Innovation Institute (TII), a leading AI research company based in the UAE, in September 2023. It had 180 billion parameters and was trained on 3.5 trillion tokens from TII's RefinedWeb dataset, which contained high-quality web pages from various domains. Falcon 180B could be used for various natural language tasks such as chat, instruction, and generation, and was accessible on the Hugging Face platform. Falcon on open source, available for both research and commercial purposes.

Mistral Co-founded by alumni from Google's DeepMind and Meta, Mistral AI was a Paris-based startup focused on foundational models with an open technology approach. In December 2023, it closed a \$415 million Series A round led by Andreessen Horowitz, valuing the company at approximately \$2 billion. Mistral AI released its first model, Mistral 7B (trained on a data set of 7 billion parameters), as a free download under the Apache 2.0 license, emphasizing an open-source approach rather than direct competition with models like GPT-4 or Claude 2. The company played a significant role in the EU's AI Act discussions, advocating for exemptions for foundational models. Mistral AI launched a developer platform in beta, offering access to its models via APIs, including the new Mistral 8x7B model, which used a router network for efficient parameter processing. In February 2024, Mistral released Mistral Large, its flagship LLM, which was not open source and could be accessed through Mistral's own API platform, potentially rivaling GPT-4 or Claude 3. Microsoft invested a further \$16M in Mistral AI, and forged a distribution deal for Azure, allowing Azure customers to access Mistral's models through Azure's model catalog.

Source: Compiled by casewriters from Cade Metz, "Google Releases Gemini, an A.I.-Driven Chatbot and Voice Assistant," *The New York Times*, February 8, 2024, <https://www.nytimes.com/2024/02/08/technology/google-gemini-ai-app.html>, accessed March 2024; CB Insights, Large Language Model Operations Market Map, October 2023; Shikhar Ghosh and Shweta Bagai, "OpenAI: Idealism Meets Capitalism," Harvard Business School Case 824-134, January 2024; Andy Wu and others, "AI Wars," Harvard Business School Case 723-434, June 2023; Shritama Saha, "UAE Unveils GPT-4's New Rival, Falcon 180B," *AIM*, September 7, 2023, <https://analyticsindiamag.com/uaes-tii-unveils-falcon-180b-a-new-open-source-llm/>, accessed December 2023; Romain Dillet, "Mistral AI, a Paris-Based OpenAI rival, closed its \$415 million funding round," *TechCrunch*, December 11, 2023, <https://techcrunch.com/2023/12/11/mistral-ai-a-paris-based-openai-rival-closed-its-415-million-funding-round/>, accessed March 2024; Julia Black, "Behind the Marriage of Microsoft and Mustafa Suleyman, a Bid for redemption," *The Information*, April 6, 2024, <https://www.theinformation.com/articles/behind-the-marriage-of-microsoft-and-mustafa-suleyman-a-bid-for-redemption?rc=kimdxv>, accessed April 2024; Meta, "Introducing Meta LLaMA 3," <https://ai.meta.com/blog/meta-llama-3/>, April 18, 2024, accessed April 2024; Reuters, "Baidu says AI chatbot 'Ernie Bot' has attracted 200 million users," <https://www.reuters.com/technology/baidu-says-ai-chatbot-ernie-bot-has-amassed-200-million-users-2024-04-16/>, accessed April 2024; Romain Dillet, "Microsoft made a \$16M investment in Mistral AI," *TechCrunch*, February 27, 2024, <https://techcrunch.com/2024/02/27/microsoft-made-a-16-million-investment-in-mistral-ai/>, accessed April 2024.

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